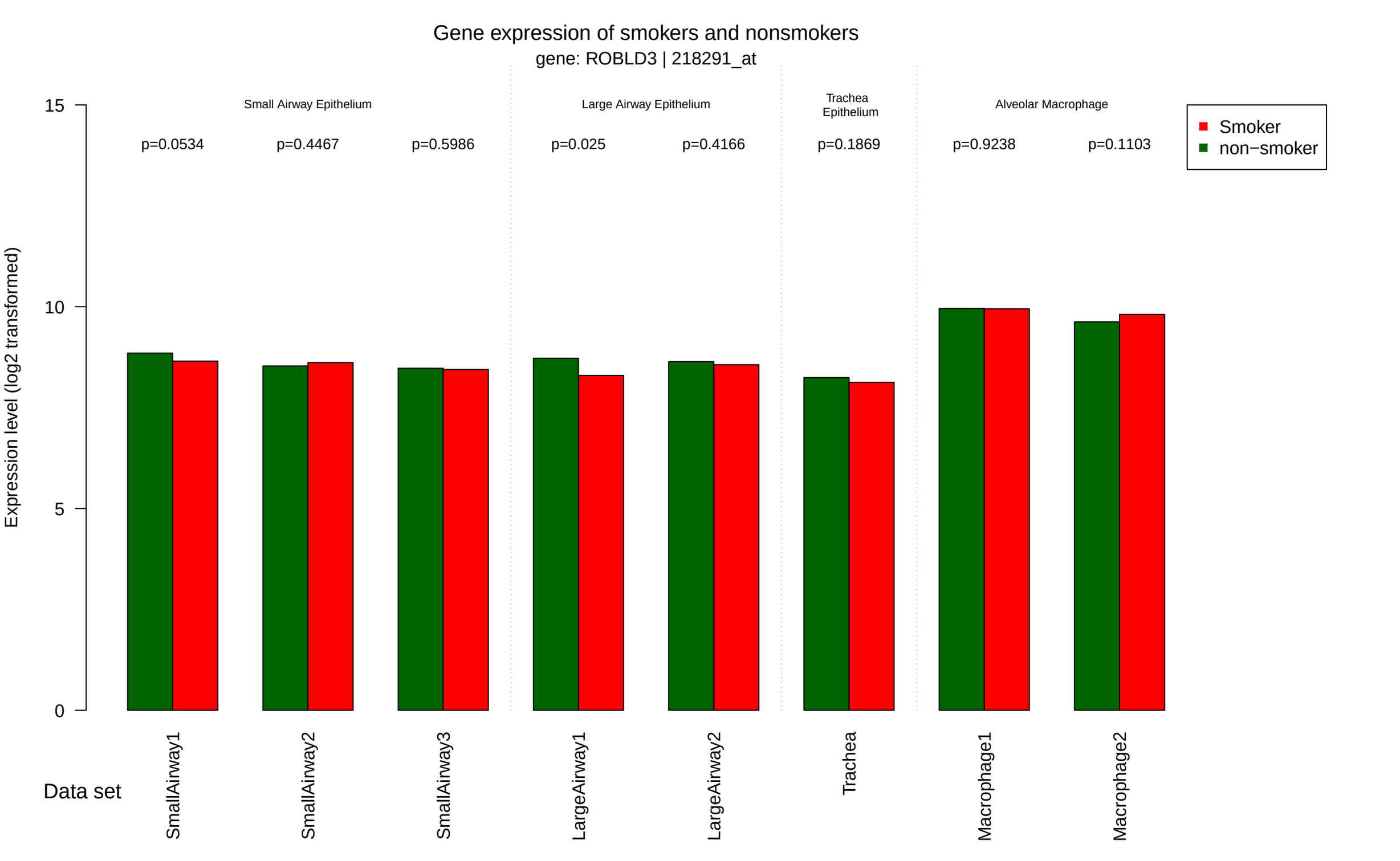
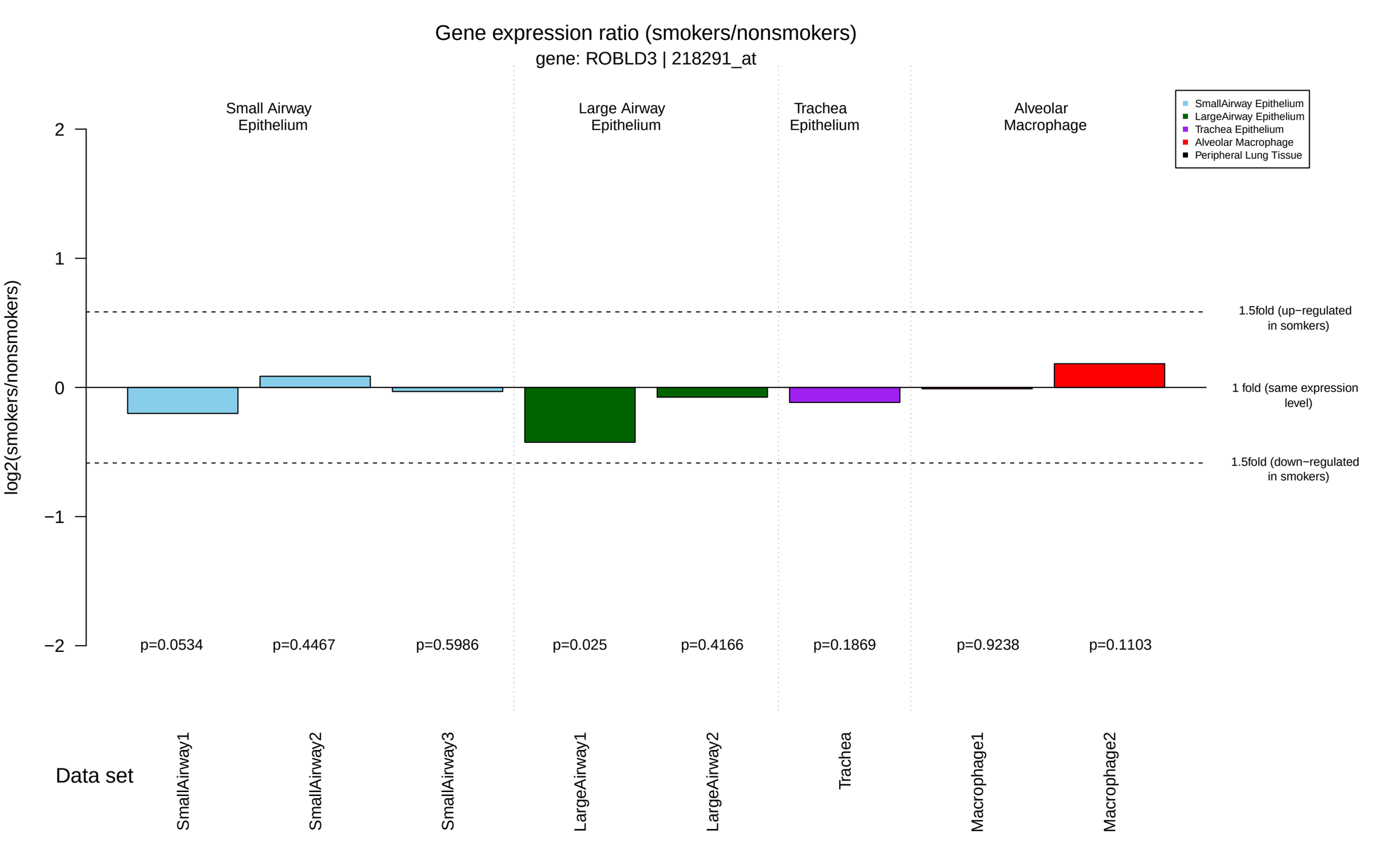
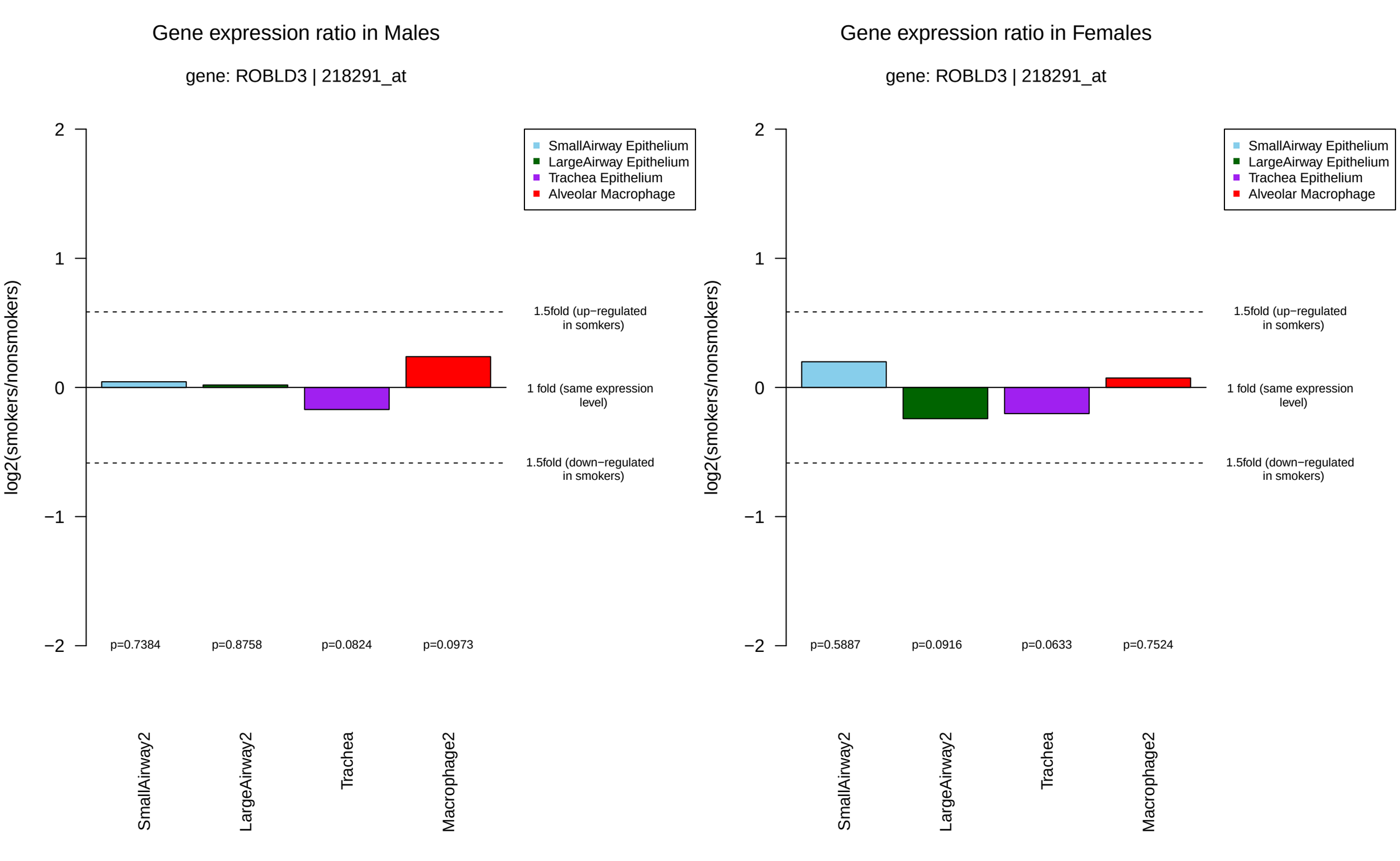
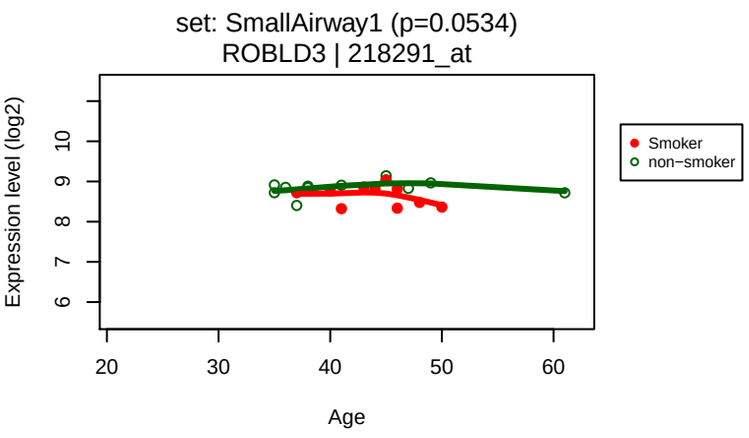
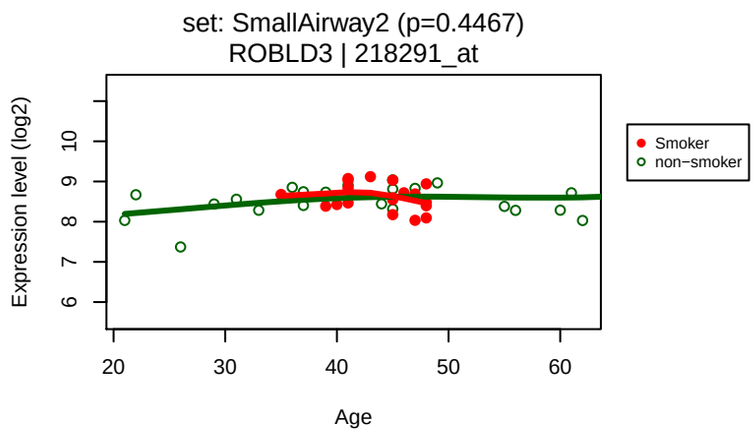
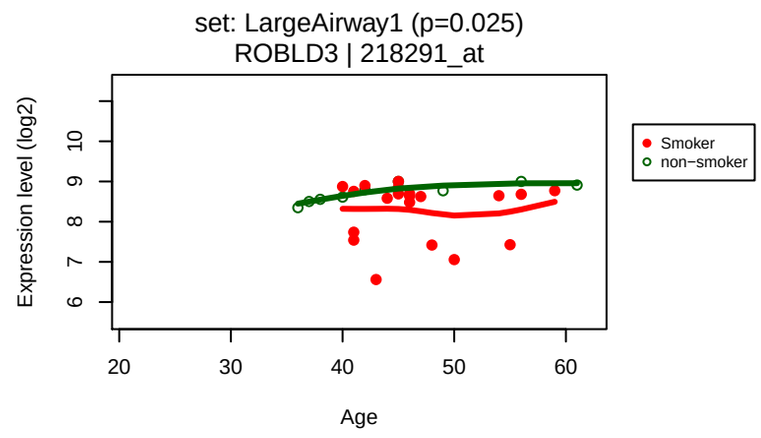

Gene expression of smokers and nonsmokers
gene: ROBLD3 | 218291_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0534
p=0.4467
p=0.5986
p=0.025
p=0.4166
p=0.1869
p=0.9238
p=0.1103
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

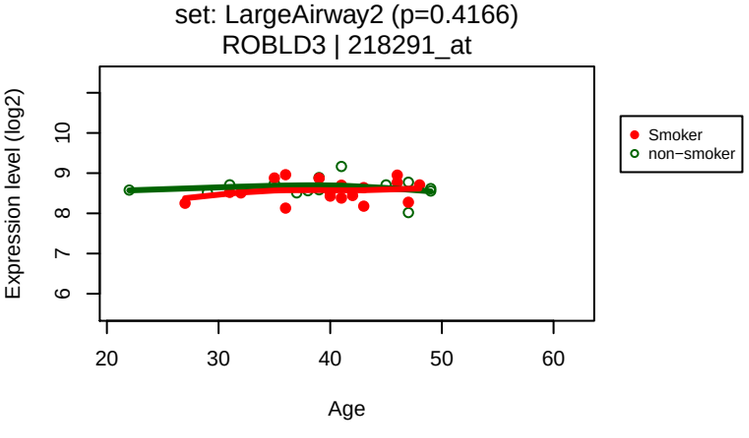

Gene expression ratio (smokers/nonsmokers)
gene: ROBLD3 | 218291_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0534
p=0.4467
p=0.5986
p=0.025
p=0.4166
p=0.1869
p=0.9238
p=0.1103
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

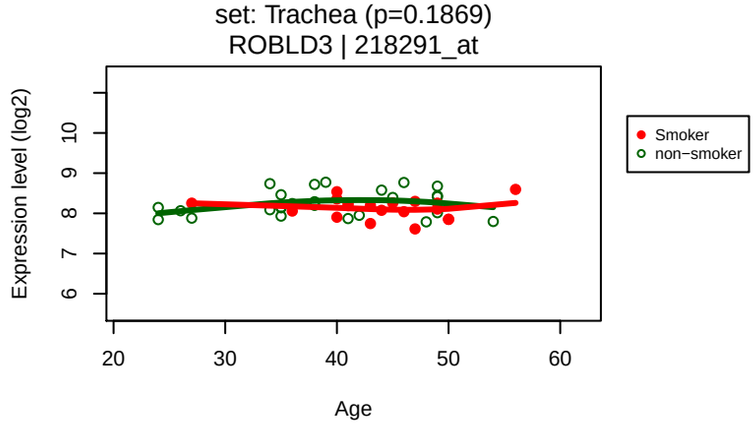

Gene expression ratio in Males
gene: ROBLD3 | 218291_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7384
p=0.8758
p=0.0824
p=0.0973
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ROBLD3 | 218291_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5887
p=0.0916
p=0.0633
p=0.7524
SmallAirway2
LargeAirway2
Trachea
Macrophage2

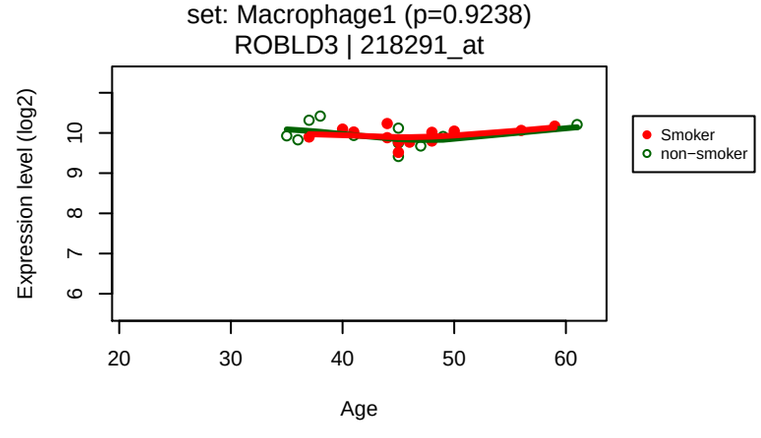

set: SmallAirway1 (p=0.0534)
ROBLD3 | 218291_at
Expression level (log2)
Age
Smoker
non−smoker

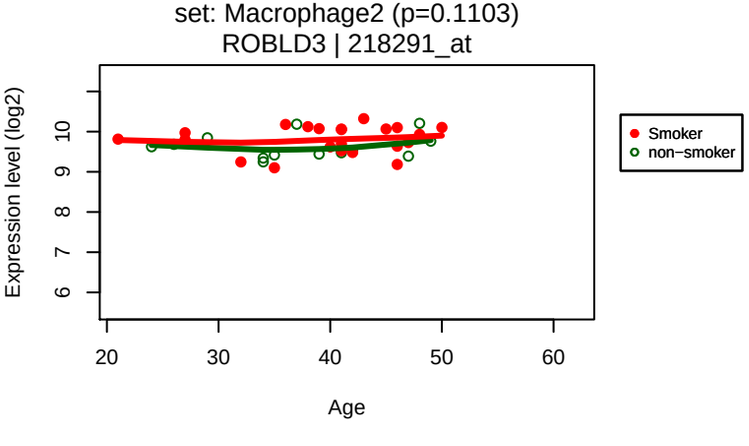

set: SmallAirway2 (p=0.4467)
ROBLD3 | 218291_at
Expression level (log2)
Age
Smoker
non−smoker

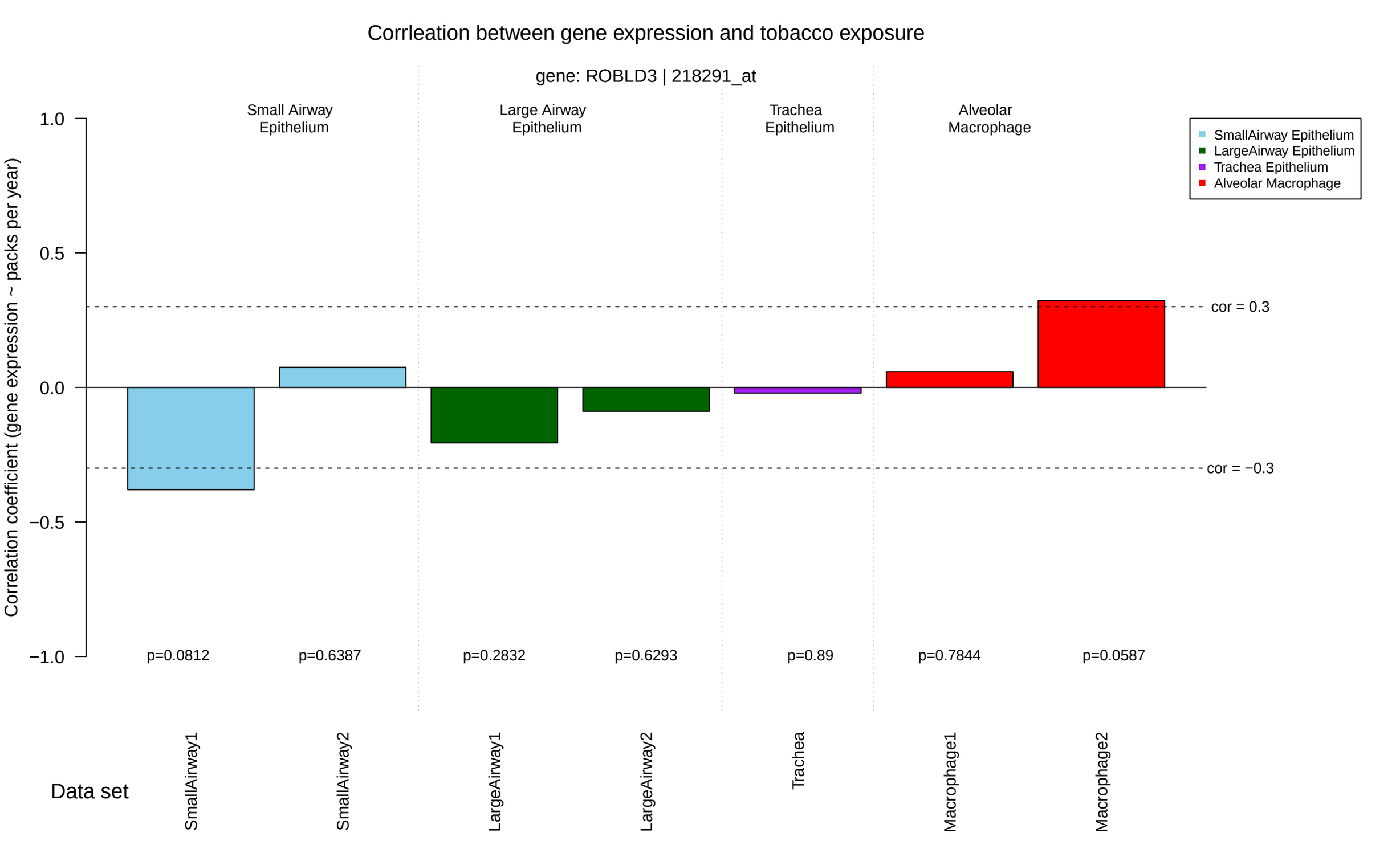

set: LargeAirway1 (p=0.025)
ROBLD3 | 218291_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4166)
ROBLD3 | 218291_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1869)
ROBLD3 | 218291_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9238)
ROBLD3 | 218291_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1103)
ROBLD3 | 218291_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ROBLD3 | 218291_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0812
p=0.6387
p=0.2832
p=0.6293
p=0.89
p=0.7844
p=0.0587
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2